Supplemental Material: Running Coupling $\mu(a) = H(a)/c$ as the Unique Power Preserving CDM-like Background Scaling

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The Khronon kinetic scale μ is, on the cosmological background, a function of the FRW scale factor. We prove that among power-law ansätze $\mu(a) = \mu_0[H(a)/H_0]^n$, only $n = 1$ preserves $\delta(a) = Q(a) - 1 =$ constant in the matter era, which is the unique behavior compatible with $\rho_K(a) \propto a^{-3}$ (CDM-like background scaling). All other powers force either (i) a runaway growth or decay of δ that is excluded by CMB constraints on $\Omega_{\rm DM}(z)$, or (ii) a fine-tuned initial condition at recombination that re-introduces $\Omega_{\rm m}h^2$ as a free parameter independent of δ_0. We solve the linearized Khronon equation of motion with Hubble friction and show that $n = 1$ is the only attractor solution. We acknowledge that the value of $\delta(a_{\rm rec})$ remains a free initial condition—the same status it has in ΛCDM via $\Omega_{\rm m}h^2$.

I. SETUP

[KNOWN] The Khronon kinetic energy density on the FRW background is [?]

$$\rho_K(a) = \frac{c^2\mu^2(a)}{8\pi G}\,\delta(a)\big[2+\delta(a)\big], \tag{1}$$

where $\delta(a) = Q(a) - 1$ and $\mu(a)$ is the kinetic scale, in general scale-factor dependent. We parametrize the running coupling as

$$\mu(a) \;=\; \mu_0\,\big[H(a)/H_0\big]^n, \qquad \mu_0 = H_0/c, \tag{2}$$

with $n \in \mathbb{R}$ to be determined. Paper 3 adopts $n = 1$ in Eq. (11), here labelled (**??**) for convenience. This supplement derives that choice from a CDM-compatibility requirement.

The matter-era Hubble rate is $H(a) = H_0\,\Omega_{\rm m}^{1/2}\,a^{-3/2}$ for $a \ll a_{\rm eq}^{(\Lambda)}$, so $\mu^2(a) \propto a^{-3n}$ in that regime.

II. THE CDM-COMPATIBILITY REQUIREMENT

[DERIVED] Pressureless dark matter dilutes as $\rho \propto a^{-3}$. Planck 2018 constrains $\Omega_{\rm DM}(z)$ at the percent level via the CMB acoustic peaks at $z \sim 1100$ and via late-time growth [?]; any deviation $\rho_K(a) \propto a^{-3+\epsilon}$ with $|\epsilon| \gtrsim 0.05$ is excluded by the combined CMB+BAO+BBN data set.

Writing $\rho_K(a) \propto \mu^2(a)\,\delta(a)[2+\delta(a)]$, the matter-era requirement $\rho_K \propto a^{-3}$ becomes

$$\mu^2(a)\,\delta(a)\big[2+\delta(a)\big] \;\propto\; a^{-3}. \tag{3}$$

Substituting $\mu^2 \propto a^{-3n}$:

$$\delta(a)\big[2+\delta(a)\big] \;\propto\; a^{-3+3n} \;=\; a^{3(n-1)}. \tag{4}$$

We now consider three cases.

A. Case $n = 1$: $\delta =$ const

For $n = 1$, Eq. (**??**) demands $\delta(a)[2+\delta(a)] =$ constant, i.e. $\delta =$ constant. This is consistent with the present-day value $\delta_0 = 0.34$ being maintained throughout the matter era. ρ_K then scales exactly as a^{-3}, identical to CDM at the background level.

B. Case $n = 0$: constant μ

For $n = 0$, $\mu = \mu_0$ is fixed. The CDM requirement becomes $\delta(2+\delta) \propto a^{-3}$, which for small δ reduces to $\delta \propto a^{-3}/2$. This forces $\delta(a_{\rm rec})$ to be larger than δ_0 by a factor $(a_0/a_{\rm rec})^3 \approx 10^9$; with $\delta_0 = 0.34$ today, $\delta_{\rm rec} \sim 3.4 \times 10^8$, well outside the small-δ regime where $K = \mu^2\delta^2$ remains a faithful description of the kinetic function. Moreover, such large δ violates the ghost condensation expansion entirely. $n = 0$ is excluded.

C. Case $n \neq 0, 1$

For general n, $\delta(a) \propto a^{3(n-1)/2}$ in the small-δ regime. Demanding $\delta(a_0) = \delta_0$ fixes the proportionality constant; the value at recombination is

$$\delta_{\rm rec} = \delta_0\,(a_0/a_{\rm rec})^{3(1-n)/2} \sim 10^{4.5(1-n)}\,\delta_0. \tag{5}$$

For $n = 1/2$: $\delta_{\rm rec} \sim 10^{2.3}\,\delta_0 \approx 70$, again outside the ghost-condensation small-δ regime. For $n = 2$: $\delta_{\rm rec} \sim 10^{-4.5}\,\delta_0 \approx 10^{-5}$, requiring an absurdly fine-tuned initial condition. Only $n = 1$ avoids any dependence of δ on the cosmological expansion.

Proposition 1. *Among power-law running couplings $\mu(a) = \mu_0[H(a)/H_0]^n$, the requirement $\rho_K(a) \propto a^{-3}$ in the matter era with $\delta(a)$ remaining in the ghost-condensation small-displacement regime ($|\delta| \lesssim O(1)$) selects $n = 1$ uniquely.*

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III. LINEARIZED KHRONON PERTURBATION

[DERIVED] A more dynamical argument starts from the Khronon equation of motion. Variation of $S_K = \int \sqrt{-g}\, K(Q)\, d^4x$ on FRW gives, to leading order in δ,

$$\ddot{\delta} + 3H\dot{\delta} + [c_s^2\, k^2/a^2 + m_{\rm eff}^2]\, \delta = 0, \tag{6}$$

where for $K = \mu^2(Q-1)^2$ the effective mass is $m_{\rm eff}^2 = 0$ and $c_s^2 = 0$ at the condensate point (Paper 3, Sec. III). In the homogeneous limit $k = 0$:

$$\ddot{\delta} + 3H\dot{\delta} = 0 \implies \dot{\delta} \propto a^{-3}, \tag{7}$$

giving $\delta = \delta_\infty + C \int a^{-3}\, dt$. The transient term decays as $a^{-3/2}$ in the matter era and is negligible after one Hubble time, so $\delta \to \delta_\infty$ asymptotically. This is the $n = 1$ attractor.

Effect of running coupling. Allowing $\mu(a) = \mu_0[H(a)/H_0]^n$ modifies (**??**) by a time-dependent normalization of the kinetic term:

$$\frac{d}{dt}\big[\mu^2(a)\, a^3\, \dot{\delta}\big] = 0, \tag{8}$$

which is the conservation law for the Khronon current (Paper 3, Sec. III, Theorem 1). Integrating: $\dot{\delta} \propto \mu^{-2} a^{-3}$. For $\mu^2 \propto a^{-3n}$: $\dot{\delta} \propto a^{-3(1-n)}$.

The energy density $\rho_K \propto \mu^2\, \delta^2$ (small-δ) then evolves as

$$\rho_K(a) \;\propto\; a^{-3n}\, \delta^2(a), \tag{9}$$

and the demand $\rho_K \propto a^{-3}$ recovers $\delta^2 \propto a^{-3+3n} = a^{3(n-1)}$, identical to Eq. (**??**) above. The Hubble-friction analysis therefore confirms the algebraic result: $n = 1$ is the unique attractor.

IV. NUMERICAL ILLUSTRATION

We solve (**??**) numerically in the matter era for $n \in \{0, 1/2, 1, 3/2, 2\}$ with the initial condition $\delta(a = 10^{-3}) = \delta_0$ and $\dot{\delta}(a = 10^{-3}) = 0$, and report the value of $\delta(a = 1)$ together with the implied $\rho_K(a)/a^{-3}$ ratio. Table **??** summarizes the results.

TABLE I. Power-law running coupling $\mu \propto H^n$. Last column shows the deviation of $\rho_K(a)$ from CDM scaling at $z = 0$, anchored to $\rho_K(z = 1100) = \rho_{\rm CDM}(z = 1100)$. A negative sign indicates ρ_K falls below CDM at late times.

n	$\delta(z=0)/\delta_{\rm rec}$	$\rho_K(z{=}0)/\rho_{\rm CDM}(z{=}0)$
0	$\sim 10^{-9}$	$\sim 10^{-18}$ (excluded)
1/2	$\sim 10^{-2}$	$\sim 10^{-2}$ (excluded)
1	**1**	**1** (CDM exact)
3/2	$\sim 10^{2}$	ghost cond. broken
2	$\sim 10^{4.5}$	ghost cond. broken

The $n = 1$ result is exact (modulo the $\delta(2+\delta)$ correction discussed in Paper 3, Sec. III.E). All other powers either exclude the model on background grounds or break the ghost-condensation regime.

V. LIMITATIONS

Initial condition. The argument above fixes the *scaling* of $\delta(a)$ but not its *value*. The amplitude $\delta(a_{\rm rec})$ remains a free initial condition—equivalent in information content to specifying $\Omega_{\rm DM}h^2$ in ΛCDM. No theory derives $\Omega_{\rm DM}h^2$ from first principles; the Khronon framework is no different in this respect. The improvement is conceptual: in ΛCDM the parameter is a particle abundance from unknown thermal-history physics; here it is a boundary condition on a classical scalar field.

Radiation era. In the radiation era, $H^2 \propto a^{-4}$, and the running coupling gives $\mu^2 \propto a^{-4n}$. The CDM requirement becomes $\delta(2+\delta) \propto a^{-3+4n}$. For $n = 1$: $\delta \propto a^{1/2}$ in the small-δ regime, growing modestly between BBN and recombination. This is consistent with $\delta_{\rm rec} \approx \delta_0/\Omega_{\rm m}$ quoted in Paper 3, Sec. III.E and is well within the small-displacement regime.

Λ era. At late times when Λ dominates, $H \to$ constant, so $\mu \to \mu_0$ asymptotically. δ is already at the attractor δ_∞ and remains there. No additional tuning is required.

Caveat on power-law ansatz. Beyond power-laws, one could consider $\mu(a) = f(H(a)/H_0)$ for arbitrary f. Within power-laws, $n = 1$ is unique; within general f, the same conclusion holds for the matter-era attractor (the Hubble-friction analysis of Sec. **??** forces $f(H) \propto H$ in any case where δ is to remain at a fixed value).

VI. CONNECTION TO PAPER 1

[NEW SYNTHESIS] The Petz recovery framework [**?**] identifies the IR scale of the Khronon as the retrodictability horizon $H(a)^{-1}c$. At any cosmological epoch, signals propagating beyond this horizon are *currently* unrecoverable; the Khronon kinetic scale tracks this evolving horizon. The choice $\mu(a) = H(a)/c$ is therefore not an auxiliary phenomenological assumption but a direct consequence of the IR structure of retrodictability: the scale at which $K(Q)$ activates is the scale at which information loss becomes order unity, and that scale runs with $H(a)$.

The constancy of $\delta = Q - 1$ in the matter era is then a statement about the constancy of the *retrodictability deficit*: the universe's "arrow strength" is a stationary quantity in the matter era, set once at recombination by the boundary condition on δ, and modified only by the radiation-matter and matter-Λ transitions.

VII. SUMMARY

The running coupling $\mu(a) = H(a)/c$ is selected by two converging arguments: (i) algebraic, demanding $\rho_K(a) \propto a^{-3}$ in the matter era forces $n = 1$ in $\mu \propto H^n$; (ii) dynamical, the Khronon equation of motion with Hubble friction has $\delta =$ constant as its unique attractor only when $n = 1$. The amplitude $\delta(a_{\rm rec})$ remains a free initial condition, equivalent to $\Omega_{\rm DM}h^2$ in ΛCDM. The improvement is that $\Omega_{\rm DM} = \delta_0(2 + \delta_0)/3$ is *computed* from δ_0 rather than fitted as a particle abundance.

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